

OSSP (Operating Systems And Systems Programming)

Course code: 25CS2104E

Section: 3

Project: University Laboratory Management Server

Project Output Screenshots:

Terminal-1

```
sushma_sree@LAPTOP-6PJSRT08:~$ cd project
sushma_sree@LAPTOP-6PJSRT08:~/project$ ls
Makefile  client.c  lab_client  lab_server  server.c  thread_test
a.out     common.h  lab_client.c  lab_server.c  server.log  thread_test.c
```

Compile the server:

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ gcc lab_server.c -o lab_server -pthread
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

Run the server:

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_server
[WORKER 1] Started.
[WORKER 2] Started.
[WORKER 3] Started.
[SERVER] Socket created successfully.
[SERVER] Server bound to port 8080.
[SERVER] Server is listening on port 8080.
[SERVER] Indian Time: 20:14 23-09-2026 IST
```

Terminal-2

Compile the client:

```
sushma_sree@LAPTOP-6PJSRT08:~$ cd project
sushma_sree@LAPTOP-6PJSRT08:~/project$ gcc lab_client.c -o lab_client
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

Run the client:

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client
```

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====
```

```
Enter Student ID: |
```

```
Enter Student ID: 101
```

```
----- MENU -----
1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status
-----
```

```
Enter your choice: 1
```

```
Enter required usage time (in seconds): 10
```

```
=====
SERVER RESPONSE
=====
```

```
=====
RESOURCE ALLOCATED
=====
```

```
Student ID    : 101
Resource      : Computer
Duration      : 10 seconds
Start Time    : 20:17 23-09-2026 IST
Expected End  : 20:17 23-09-2026 IST
-----
```

```
The resource will be automatically
released after 10 seconds.
=====
```

Checking the resource dashboard:

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====
```

Enter Student ID: 101

```
----- MENU -----
```

1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status

Enter your choice: 4

```
=====
SERVER RESPONSE
=====
```

```
=====
LAB RESOURCE DASHBOARD
=====
```

```
Computers      : 4/5 available
Printers       : 2/2 available
File Servers   : 2/2 available
=====
```

Show Student-Specific Status

```

sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client

=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====

Enter Student ID: 101

----- MENU -----
1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status
-----

Enter your choice: 6

=====
SERVER RESPONSE
=====

=====
STUDENT RESOURCE STATUS
=====
Student ID : 101

Resource      : Computer
Requested Time : 30 seconds
Used Time     : 7 seconds
Remaining Time : 23 seconds
Start Time    : 20:22 23-09-2026 IST
Expected End  : 20:23 23-09-2026 IST
=====

```

Request queue

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client
```

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====
```

```
Enter Student ID: 101
```

```
----- MENU -----
```

1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status

```
-----
```

```
Enter your choice: 2
```

```
Enter required usage time (in seconds): 10
```

```
=====
SERVER RESPONSE
=====
```

```
=====
RESOURCE ALLOCATED
=====
```

```
Student ID   : 101
Resource     : Printer
Duration     : 10 seconds
Start Time   : 20:26 23-09-2026 IST
Expected End : 20:27 23-09-2026 IST
=====
```

```
The resource will be automatically
released after 10 seconds.
```

```
=====
```

Semaphore

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client
```

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====
```

Enter Student ID: 102

----- MENU -----

1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status

Enter your choice: 2

Enter required usage time (in seconds): 20

```
=====
SERVER RESPONSE
=====
```

```
=====
RESOURCE ALLOCATED
=====
```

```
Student ID   : 102
Resource     : Printer
Duration     : 20 seconds
Start Time   : 20:28 23-09-2026 IST
Expected End : 20:28 23-09-2026 IST
```

```
-----
The resource will be automatically
released after 20 seconds.
=====
```

Allocate the second printer(semaphore)

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client
```

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====

Enter Student ID: 103

----- MENU -----
1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status
-----

Enter your choice: 2

Enter required usage time (in seconds): 60

=====
SERVER RESPONSE
=====

=====
RESOURCE ALLOCATED
=====
Student ID    : 103
Resource      : Printer
Duration      : 60 seconds
Start Time    : 20:30 23-09-2026 IST
Expected End  : 20:31 23-09-2026 IST
-----

The resource will be automatically
released after 60 seconds.
=====
```

Request for 3printer

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ./lab_client
```

```
=====
UNIVERSITY LAB MANAGEMENT CLIENT
=====

Enter Student ID: 104

----- MENU -----
1. Request Computer
2. Request Printer
3. Request File Server
4. View Resource Dashboard
5. View All Allocated Resources
6. View My Resource Status
-----

Enter your choice: 2

Enter required usage time (in seconds): 90
|
```

In server we get as,

```
[SERVER] Request received from Student 104.
[SERVER] Request queued for Student 104.
[WORKER 1] Processing Student 104 request.
[RESOURCE] Student 104 waiting for Printer.
```

Get the Server PID

Terminal-3

```
sushma_sree@LAPTOP-6PJSRT08:~$ pgrep -a lab_server
517 ./lab_server
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

SIGUSR1 (it shows number of pending requests)

```
sushma_sree@LAPTOP-6PJSRT08:~$ kill -USR1 517
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

In server we get,

```
[SIGNAL] Pending requests: 0
```

SIGUSR2 (to check resource availability)

```
sushma_sree@LAPTOP-6PJSRT08:~$ kill -USR2 517
```

In server we get,

```
[SIGNAL] Resource status requested.
Computers      : 5/5 available
Printers       : 2/2 available
File Servers   : 2/2 available
|
```

To show the threads

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ps -T -p 517
  PID   SPID TTY      TIME CMD
   517    517 pts/0    00:00:00 lab_server
   517    518 pts/0    00:00:00 lab_server
   517    519 pts/0    00:00:00 lab_server
   517    520 pts/0    00:00:00 lab_server
   517    521 pts/0    00:00:00 lab_server
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

Server activity log(gives the entire records)

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ cat server.log
SERVER STARTED
[17-09-2026 05:23:53] =====
[17-09-2026 05:23:53] SERVER STARTED
[17-09-2026 05:23:53] Worker 1 started.
[17-09-2026 05:23:53] Worker 2 started.
[17-09-2026 05:23:53] Worker 3 started.
[17-09-2026 06:38:51] =====
[17-09-2026 06:38:51] SERVER STARTED
[17-09-2026 06:38:51] Worker 2 started.
[17-09-2026 06:38:51] Worker 1 started.
[17-09-2026 06:38:51] Worker 3 started.
[17-09-2026 06:38:51] Server bound to port 8080 and waiting for clients.
[17-09-2026 06:39:03] Shutdown signal received.
[17-09-2026 06:39:03] Worker 2 stopped.
[17-09-2026 06:39:03] Worker 1 stopped.
[17-09-2026 06:39:03] Worker 3 stopped.
[17-09-2026 06:39:03] Server shutdown started.
[17-09-2026 06:39:03] SERVER STOPPED
[17-09-2026 06:39:03] =====
[17-09-2026 06:41:52] =====
[17-09-2026 06:41:52] SERVER STARTED
[17-09-2026 06:41:52] Worker 1 started.
[17-09-2026 06:41:52] Worker 2 started.
[17-09-2026 06:41:52] Worker 3 started.
[17-09-2026 06:41:52] Server bound to port 8080 and waiting for clients.
[17-09-2026 06:41:52] Server is listening for client connections.
```

controlled shutdown

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ kill -TERM 517
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

In server,

```
[SIGNAL] Shutdown signal received.
[WORKER 1] Stopped.
[WORKER 3] Stopped.
[WORKER 2] Stopped.
[SERVER] Server stopped safely.
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

Verify that the server has stopped

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ pgrep -a lab_server
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```

If the server stopped successfully, nothing should be displayed.

Final verification of the project files:

```
sushma_sree@LAPTOP-6PJSRT08:~/project$ ls -l lab_server.c lab_client.c lab_server lab_client server.log
-rwxr-xr-x 1 sushma_sree sushma_sree 16640 Sep 23 14:46 lab_client
-rw-r--r-- 1 sushma_sree sushma_sree  4085 Sep 18 03:56 lab_client.c
-rwxr-xr-x 1 sushma_sree sushma_sree 35712 Sep 23 14:43 lab_server
-rw-r--r-- 1 sushma_sree sushma_sree 29916 Sep 18 04:08 lab_server.c
-rw-r--r-- 1 sushma_sree sushma_sree 18792 Sep 23 15:17 server.log
sushma_sree@LAPTOP-6PJSRT08:~/project$ |
```